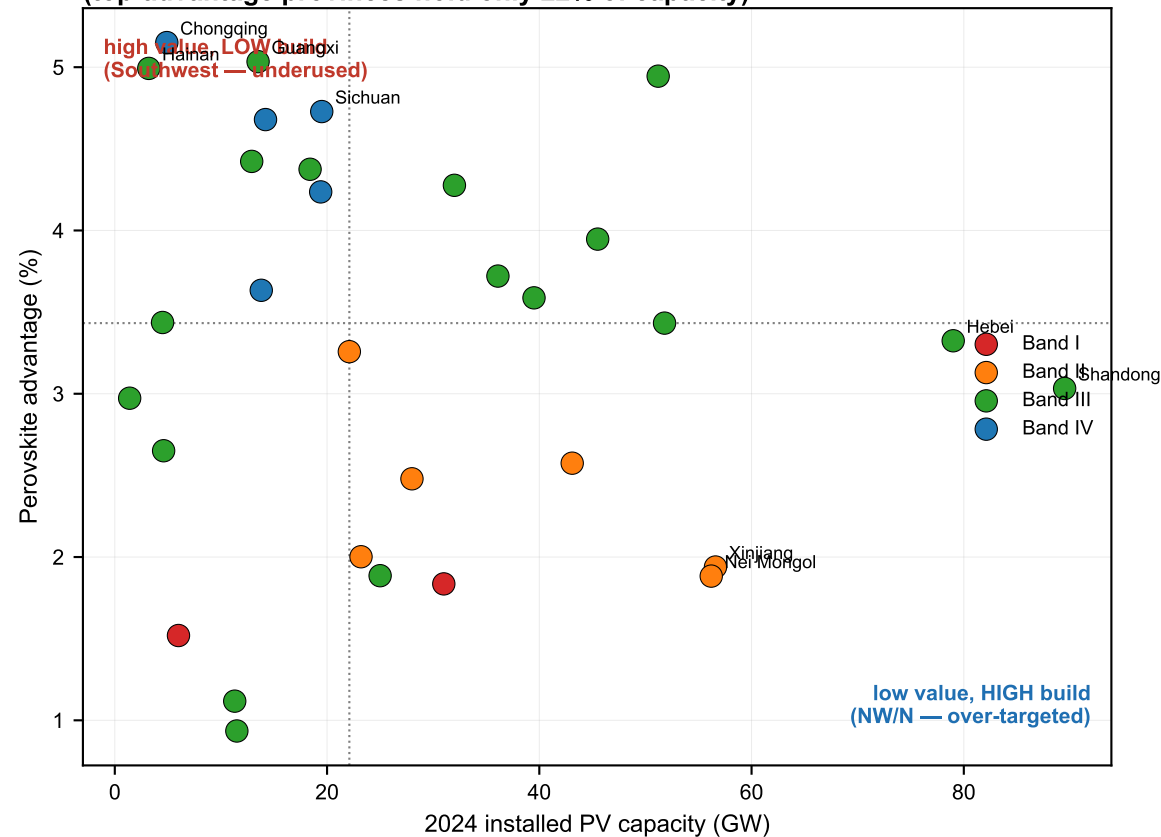
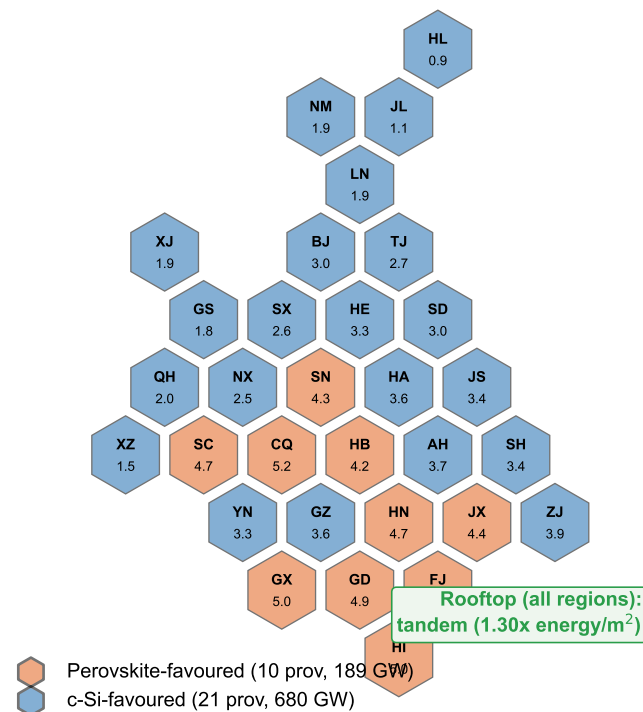


Fig 5 — Deploy by physics: tandem on rooftops, perovskite in the utility Southwest, c-Si in the Northwest; high-latitude utility also pays a land penalty

(a) Utility today: built where it matters least
(top-advantage provinces hold only 22% of capacity)



(b) Utility decision map (threshold 4.0% advantage)



(c) High-latitude utility pays a land penalty
(tech-independent: row spacing)

